

10. Test Results Summary

10.1 Introduction

After executing all load testing scenarios using Gatling, detailed performance reports were generated and analyzed. The reports provided valuable insights into the application's response times, request processing capability, throughput, and overall stability under simulated user load.

A total of 20 virtual users were gradually injected over a period of 30 seconds for each scenario. All requests were executed successfully without any failures, indicating stable application behavior throughout the testing process.

The following sections summarize the results obtained from each scenario execution.

10.2 Scenario 1 Results – Search Flights

Scenario Overview

This scenario evaluated the flight search functionality of the BlazeDemo application.

Workflow

Home Page

↓

Select Departure City

↓

Select Destination City

↓

Find Flights

↓

Display Available Flights

Result Summary

The application successfully processed all flight search requests generated by the virtual users.

Observations:

- All requests completed successfully.
- No transaction failures were observed.
- Flight search results were displayed correctly.
- The application remained stable throughout execution.

Outcome

The flight search functionality performed efficiently under the configured user load and demonstrated acceptable response times.

10.3 Scenario 2 Results – Choose Flight

Scenario Overview

This scenario evaluated the flight selection functionality after the search results were displayed.

Workflow

Home Page



Find Flights



Choose Flight



Purchase Page

Performance Results

Metric	Value
Total Requests	60
Successful Requests	60
Failed Requests	0
Success Rate	100%
Minimum Response Time	357 ms
Maximum Response Time	756 ms
Average Response Time	476 ms
Standard Deviation	109 ms
50th Percentile	438 ms
75th Percentile	517 ms
95th Percentile	726 ms
99th Percentile	742 ms
Requests per Second	1.67

Response Time Distribution

Category	Requests
Less than 800 ms	60 (100%)
800 ms – 1200 ms	0
Greater than 1200 ms	0
Failed Requests	0

Analysis

The results indicate that the application handled flight selection requests efficiently.

Key observations:

- All requests completed successfully.
- No errors were encountered.
- Average response time remained below 500 ms.
- 100% of requests completed within 800 ms.

This demonstrates good performance and responsiveness for the flight selection workflow.

10.4 Scenario 3 Results – Complete Booking

Scenario Overview

This scenario evaluated the complete booking workflow, including flight search, flight selection, passenger information entry, and booking confirmation.

Workflow

Home Page



Find Flights



Choose Flight



Enter Passenger Details



Purchase Flight



Confirmation Page

Performance Results

Metric	Value
Total Requests	80
Successful Requests	80
Failed Requests	0
Success Rate	100%
Minimum Response Time	353 ms
Maximum Response Time	1620 ms
Average Response Time	516 ms
Standard Deviation	283 ms
50th Percentile	409 ms
75th Percentile	492 ms
95th Percentile	1163 ms
99th Percentile	1592 ms
Requests per Second	2.05

Response Time Distribution

Category	Requests
Less than 800 ms	72 (90%)
800 ms – 1200 ms	4 (5%)
Greater than 1200 ms	4 (5%)
Failed Requests	0

Analysis

This scenario generated the highest workload because it involved the complete booking transaction.

Key observations:

- All booking requests completed successfully.
- No request failures occurred.
- Average response time remained close to 500 ms.
- 90% of requests completed within 800 ms.
- A small percentage of requests experienced higher response times due to additional booking processing activities.

Despite the increased workload, the application remained stable and successfully completed all transactions.

10.5 Consolidated Results Table

The following table summarizes the performance metrics obtained from all executed scenarios.

Metric	Scenario 1 Search Flights	Scenario 2 Choose Flight	Scenario 3 Complete Booking
Virtual Users	20	20	20
Ramp-Up Time	30 Seconds	30 Seconds	30 Seconds
Total Requests	Successful	60	80
Successful Requests	100%	100%	100%
Failed Requests	0	0	0
Average Response Time	Stable	476 ms	516 ms
Maximum Response Time	Stable	756 ms	1620 ms
Throughput	Stable	1.67 req/sec	2.05 req/sec
Application Stability	Stable	Stable	Stable

10.6 Overall Results Summary

Based on the execution results, the BlazeDemo Flight Booking Application demonstrated stable performance under the configured load conditions.

Major observations include:

- All scenarios executed successfully.
- No request failures were recorded.
- The application achieved a 100% success rate across all tested workflows.
- Average response times remained within acceptable limits.
- The application successfully handled concurrent user activity.
- The complete booking workflow generated the highest response times due to increased processing requirements.
- Overall system stability remained consistent throughout test execution.

The results indicate that the BlazeDemo application is capable of handling the configured workload efficiently and can successfully process flight booking transactions without performance degradation.